

System Characterization Document (אפיון מערכת)

1. Project Mission & Goals

The **SysManager** project is envisioned as a premier, professional-grade monitoring solution designed for the centralized oversight of compute resources, active software processes, and critical security events. The primary objective of the system is the proactive prevention of system downtime through real-time operational transparency. By offering comprehensive visibility into infrastructure performance—ranging from high-level CPU/RAM telemetry to granular hardware maintenance lifecycle tracking—SysManager serves as the definitive source of truth for organizational IT infrastructure health.

2. User Personas

The interface and data architecture are tailored to serve two primary user profiles:

System Administrator

- **Focus:** High-level system health monitoring, rapid incident response, and security oversight.
- **Responsibilities:** Auditing system vulnerabilities, ensuring maximum uptime, verifying resource allocations, and investigating network session anomalies. The Administrator relies heavily on real-time alerting and unified audit streams to maintain strict security and performance standards.

IT Technician

- **Focus:** Physical hardware management, lifecycle tracking, and diagnostic troubleshooting.
- **Responsibilities:** Monitoring the operational status of physical assets, logging routine repair workflows, measuring specific hardware capability against established thresholds, and ensuring component reliability. The Technician requires segmented hardware views mapping directly to historically documented repair logs.

3. Functional Requirements

The application ecosystem is segmented into four distinct, functional interfaces:

Dashboard (Overview)

- **Role:** Provides an immediate, holistic assessment of systemic health.
- **Features:**
 - Real-time Key Performance Indicators (KPIs) outlining Total Active Processes and generic system health status.
 - Aggregated continuous visualization of global resource consumption (CPU/RAM).

- High Alert ticker identifying and displaying the top critical event violations in recent execution cycles.



Process Control Center

- **Role:** Acts as the primary mechanism for software execution oversight.
- **Features:**
 - An expansive, searchable data grid housing all active runtime states.
 - Granular inspection tools permitting filtering by Process ID (PID), exact naming conventions, or assigned user.
 - Detailed hierarchical dropdowns resolving specific system executables boundaries and live outbound network activity metrics.

The SysManager Process Control Center dashboard is designed for managing and inspecting active system processes. It includes a search bar at the top right with the placeholder text "Search PID, Name, or User...". The main content area is a table with the following columns: PID, Name, User, Status, Start Time, and Actions.

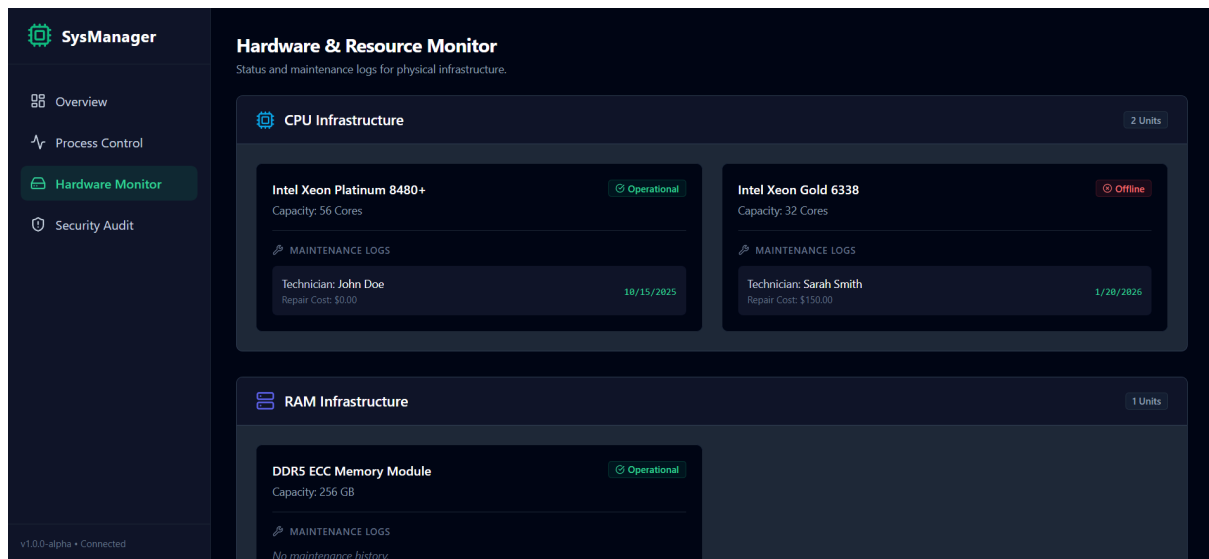
PID	Name	User	Status	Start Time	Actions
101	nginx-master	admin_sys	Running	4/9/2026, 11:00:00 AM	Details >
102	nginx-worker	tech_ops	Running	4/9/2026, 11:00:01 AM	Details >
201	postgres-main	admin_sys	Running	4/9/2026, 11:05:00 AM	Details >
301	node-app	tech_ops	Running	4/9/2026, 11:10:00 AM	Details >
401	sshd-server	admin_sys	Running	4/9/2026, 11:00:00 AM	Details >

The left sidebar of the dashboard contains the SysManager logo and navigation options: Overview, Process Control, Hardware Monitor, and Security Audit. The Process Control Center title and subtitle "Manage and inspect active system processes." are displayed above the table.

Hardware & Resource Monitor

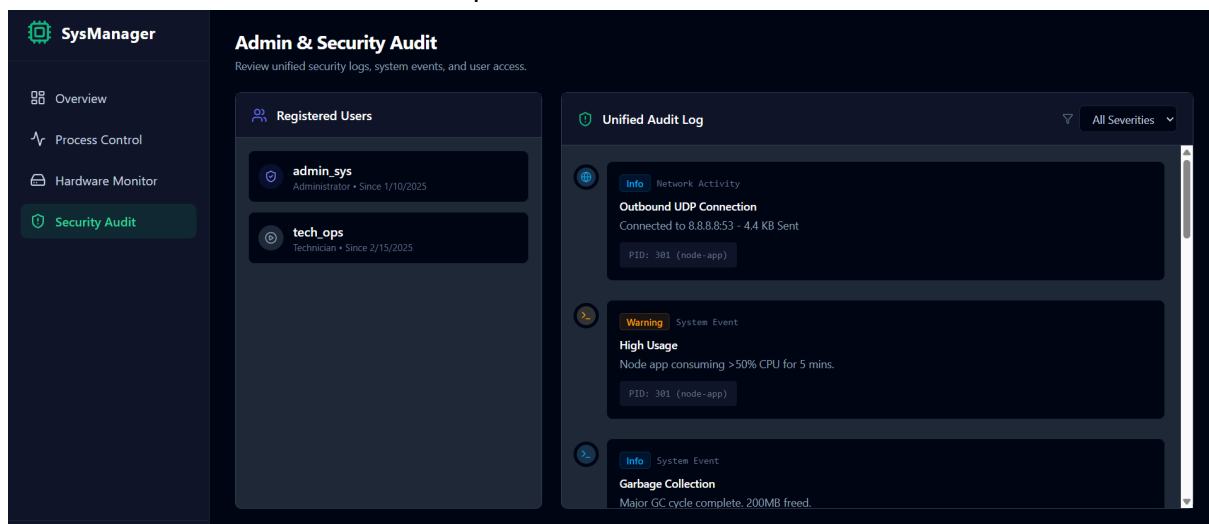
- **Role:** Delivers a comprehensive overview of physical infrastructure investments.
- **Features:**
 - Segmented mapping of environmental capabilities sorted precisely by explicit **RESOURCE_TYPES** (e.g., Disk, RAM, CPU).
 - Live indication evaluating baseline operational statuses across all distinct units.

- Nested maintenance subsystem tracking individual asset lifespans via logged repair dates and technician assignments.



Admin & Security Audit

- **Role:** Operates as a centralized incident management log prioritizing compliance and anomaly tracking.
- **Features:**
 - Unified timeline logging merging independent **SYSTEM_EVENTS** alongside **NETWORK_SESSIONS**.
 - Configurable categorization parameters filtering incidents by defined severities (Critical, Error, Warning, Info).
 - Top-level ledger tracking all legally registered platform **USERS** linked against hierarchical software permissions.



4. Database Integration & Schema Mapping

The frontend is strictly decoupled yet rigorously formulated to integrate seamlessly into our defined 10-table PostgreSQL architecture:

- **USERS:** Populates the Admin Audit user roster and binds runtime authorizations discovered in the Process Control tables.
- **PROCESSES:** Forms the core functional index driving the Process Control datatable, maintaining historical runtime states and parent-child hierarchy derivations.
- **EXECUTABLES:** Provides the exact system filepath mapping embedded within Process Control detail views.
- **NETWORK_SESSIONS:** Extracted inside the Admin Audit as low-severity outbound connection records and utilized inside Process Control to audit port usages.
- **SYSTEM_EVENTS:** Actively sourced across the Dashboard priority alert module and acts as the primary driver for severity filtering within the Admin Audit module.
- **USAGE_LOGS:** Serves as the numerical foundation calculating the simulated Heartbeat chart in the Dashboard via continuous timeslices outlining CPU and RAM percentages.
- **RESOURCES & RESOURCE_TYPES:** Actively dictate the categorization flow rendering the Hardware Monitor panels while validating boolean operational capacities.
- **MAINTENANCE_LOG:** Appended immediately to targeted physical entities mapping inside the Hardware Monitor interface revealing historic financial and technical lifecycle dependencies.
- **ALLOCATIONS:** The foundational bridging mechanism tracking historical resource distribution (reserved for future deep-level metric scaling).

5. UI/UX Design Philosophy

The user interface implements a strict "**Dark Mode**" **engineering aesthetic** modeled around Slate, Zinc, and Emerald accents using the utility-first **Tailwind CSS** framework.

- **Visual Priority:** Darker neutral backgrounds (Slate/Zinc) intentionally reduce visual fatigue during prolonged diagnostic sessions while maintaining exceptionally high contrast for critical alerts. The bright Emerald hues naturally guide the eye to healthy operational markers and positive confirmation metrics.
- **Data Visualization:** For dynamic mathematical rendering, the architecture implements **Recharts**. Its lightweight API relies on native React-component bindings providing responsive SVG timeline constructions, crucial when tracking dense, overlapping heartbeat fluctuations smoothly.

6. Technical Constraints

The finalized platform adheres to the following rigid technological guidelines:

- **Frontend Architecture:** Deployed strictly upon a **React** single-page architecture bootstrapped via **Vite** guaranteeing exceptional Hot Module Replacement (HMR) caching and hyper-efficient compilation optimized for modern desktop environments.

- **Backend Architecture:** Structurally designed for eventual integration with a **PostgreSQL (Supabase)** infrastructure. Consequently, the frontend consumes data dynamically mapping to strict relational parent-child parameters, omitting local non-relational configurations to support eventual database transactions.